

A patient-doctor relationship questionnaire (PDRQ-9) in primary care: development and psychometric evaluation

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Abstract

In health care research, the patient-doctor relationship as perceived by the patient is considered important. Our aim was to develop and validate a questionnaire that assesses the Patient-Doctor Relationship, the Patient-Doctor Relationship Questionnaire (PDRQ-9). The PDRQ-9 was developed out of the Helping Alliance Questionnaire of Luborsky, a scale that measures the therapeutic alliance in psychotherapy. Its psychometric qualities and validity in general practice were assessed, with the collaboration of 110 general practice patients and 55 patients in an Epilepsy Clinic. Principal components analysis demonstrates a two-factorial structure, one related to the doctor, and one related to the medical symptoms of the patient. Both show high reliability but as the second factor explains only 9% of the variance, it is eliminated from the questionnaire. The PDRQ-9 provides researchers a brief measure of the therapeutic aspects of the patient-doctor relationship in the primary care setting. It is a valuable tool for scientific and practical purposes involving the monitoring of the patient-doctor relationship. © 2004 Elsevier Inc. All rights reserved.

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1. Introduction

Many publications in the field of primary care mention the importance of the patient-doctor relationship. The assertion is that the ‘mental health service provision’ in primary care will improve if attention is paid to improving the patient-doctor relationship. Several issues are mentioned. First, there is a tendency to link difficult patient-doctor relationships to difficult patients [1–6].

This literature generally discusses the difficult patient-doctor relationship from the clinician’s perspective and the instruments used measure physician responses. However, at least one study has shown that both clinicians and patients experience difficult encounters as difficult, and that both physician and patient factors play a role in such encounters [7]. Hahn et al. developed the Difficult Doctor-Patient Re-

lationship Questionnaire [3] and found that “difficult” patients are prevalent in primary care settings and have more psychiatric disorders, functional impairment, health care utilization, and dissatisfaction with care. Somatoform disorder, panic disorder, dysthymia, generalized anxiety disorder, major depressive disorder, and alcohol abuse or dependence had particularly strong associations with difficulty. They suggest that studies are needed to determine whether improved diagnosis and management of mental disorders in difficult patients could diminish their excess disability, health care costs, and dissatisfaction with medical care, as well as the physicians experience of difficulty [8]. Such studies have been performed and with positive outcome [9,10].

The second issue is the communicative aspect of the patient-doctor relationship, which is studied by qualitative designs that describe the positive role of affective behavior of physicians and communication models [11–13]. In these studies the patient-doctor relationship is explored in terms of its therapeutic effect, the need to consider the patients’

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models of their illness, and the patients' expectations of their doctor. It was found that patients search out a Primary Care Practitioner (PCP) who matches their own representation of an ideal and validate their choice in a dynamic communicative process [14]. The vast literature on the quality of patient-doctor communication will not be reviewed here.

A third issue is patient satisfaction with the primary care practice, the PCP, the community nurse, or community care in general. This is assessed in a qualitative design by the Roter Interaction Analysis System (RIAS) [15]. This method has mostly been used in the primary care setting [16]. For quantitative measures, patient satisfaction scales were developed to measure patients satisfaction with care, focussing on patient involvement in decision-making [Patients' Perceived Involvement in Care Scale (PICS)] [17]. The satisfaction with consultations in primary care, reflecting the multidisciplinary nature of primary health care, is measured by the Consultation Satisfaction Questionnaire [18,19]. The Patient Satisfaction Questionnaire measures not only interaction but also technical and accessibility of services items [20]. One of the most widely used patient satisfaction questionnaires originates from the Medical Outcome Study (MOS-9) [21]. Satisfaction scales are useful for the evaluation of decision-making processes or consultations in primary care, when long-term relationships of the patient with the primary care practice are studied.

A fruitful area of research in psychotherapy has been the Helping Alliance [11]. Some aspects of the patient-doctor relationship have commonalities with this psychotherapeutic alliance, i.e., high levels of trust, helpfulness, empathic understanding and interpersonal openness. Therefore, we undertook to adapt an existing instrument from the psychotherapeutic tradition for use in primary care research. We wanted to develop a questionnaire that is short and aimed at quantification, to be used in primary care and public health research as well as in health promotion strategies. This questionnaire should focus on how well the patient regards his PCP as an effective and helpful health professional. It is known from psychotherapeutic intervention research that this is important for the effectiveness of psychotherapeutic interventions [22,23]. Alexander and Luborsky developed the Helping Alliance Questionnaire (HAQ) and Van der Linden [24] validated a Dutch version. We decided to take the Dutch version of the HAQ as a starting point for development of a short questionnaire for measuring the relationship between the PCP and the patient, from a patient point of view, and named it the Patient-Doctor Relationship Questionnaire (PDRQ). As a language check, a native speaker translated the Dutch version of the HAQ into English and back into Dutch. The aim of the present study is to validate the PDRQ in the primary care setting.

For the construct validity analysis, an additional medical setting for chronically ill patients was sought, as it was

anticipated that the patients would score their relationship with their PCP differently in that setting. A specialized neurological Epilepsy Clinic was chosen for this purpose. In that setting, the data were collected away from the primary care setting so the patients might be more honest, the patients might contrast the PCP consultations with those they experienced in the clinic and then might be less satisfied with their PCP as he had not been able to identify or treat the origin of their symptoms. This could be a way to measure the ability of the PDRQ to differentiate between a more and less critical assessment of the patient-doctor relationship.

2. Methods

2.1. Setting

The study was carried out in 3 primary care practices with 5 PCPs and in a specialized neurological Epilepsy Clinic.

2.2. Development of the PDRQ

2.2.1. Design of the questionnaire: item creation and selection

The HAQ of Luborsky contains 11 items and served as basis for item creation and selection in the PDRQ.

In the HAQ, insight into treatment is itemized (e.g., 'I gained new insight'; 'I have the feeling now that I can understand myself and can help myself'). In a patient-doctor relationship, this insight is different. It is more concerned with agreement on the treatment. Therefore, the item was rephrased as 'I benefit from the treatment of my PCP'.

Psychotherapeutics sessions are structured, with regular frequency and length. The consultations in the PCP setting are not. The following items have therefore been added: 'My PCP has enough time for me', 'My PCP is dedicated to help me'.

The HAQ contains statements that focus on the 'problems' of the patient. The underlying assumption is that if the patient thinks that his 'problems' will disappear, it indicates a well-developed psychotherapeutic relationship. For the use in the PCP setting, the 'problems' item could not be used as such, because patients usually visit their PCP for medical symptoms. Therefore, the term 'medical symptoms' was used instead of 'problems'. ['My PCP and I agree on the nature of my medical symptoms', 'I can handle my medical symptoms now (even if my PCP and I would not meet any more)']. Psychometric analysis would have to indicate whether or not these statements would remain useful in the PCP setting.

2.2.2. Pilot testing

The questionnaire was tested on 8 healthy volunteers, who were asked to fill it in, and were interviewed after-

Table 1
Mean, variance, skewness and kurtosis of the items in the questionnaire

Nr	Item PDRQ	Mean	Variance	Skewness	Kurtosis
1	My PCP understands me	3.95	.99	-.62	-.53
2	I trust my PCP	2.56	1.61	.27	-1.11
3	My PCP is dedicated to help me	3.86	1.14	-.51	-.64
4	I can talk to my PCP	2.70	1.76	.13	-1.22
5	I feel content with my PCP's treatment	2.97	1.91	.00	-1.31
6	I think my PCP finds me hard to deal with	1.41	0.77	2.46	5.99
7	My PCP helps me	4.04	1.31	-1.00	0.03
8	My PCP has enough time for me	3.81	1.25	-.60	-.61
9	I benefit from the treatment of my PCP	3.92	1.21	-.76	-.43
10	My PCP and I agree on the nature of my medical symptoms	3.80	1.55	-.69	-.58
11	I find my PCP easily accessible	3.57	1.37	-.38	-.77
12	Thanks to my PCP I feel better	3.76	1.53	-.73	-.49
13	Thanks to my PCP I gained new insight	3.93	1.25	-.78	-.29
14	I can handle my medical symptoms now (Even if my PCP and I have no further meetings)	3.87	1.26	-.73	-.264
15	My medical symptoms will probably disappear	3.41	1.44	-.39	-.66

wards. The main aim was to see if the items were comprehensible. The goal of the questionnaire was explained to them, and they were asked if they considered the statements to be relevant and comprehensive. This feedback resulted in the addition of one statement concerning accessibility of the PCP ('I find my PCP easily accessible'). Also, the appropriateness of the statements was simplified to a scale of 1 (not at all) through 5 (totally). This procedure resulted in a 15-item questionnaire.

2.3. Study sample

Five PCPs handed out PDRQs to 200 consecutive patients. In the Epilepsy Clinic, a research assistant handed out 55 PDRQs to consecutive patients. The patients filled them in at home and returned them to our department by mail. Their anonymity was guaranteed, as was the fact that their PCPs would not be informed about the results. This strategy was followed to encourage an honest assessment of their patient-doctor relationship, thus reducing response bias. Of the 255 patients, 165 (65%) responded. The respondents were predominantly Caucasian, 64% female, and had a mean age of 41.

2.4. Follow-up measurement

In the first questionnaire, patients were asked to indicate if they wanted to complete a follow-up questionnaire. Those who volunteered [35] received a second questionnaire by mail 2 months later. They all responded.

2.5. Analysis

Both skewness and kurtosis of the PDRQ items were considered. Factor analysis was performed as well as determination of Cronbach alpha and inter item correlations.

Test-retest reliability was assessed with Pearson correlations. Comparing the PDRQ scores of the primary care population with the volunteers from the Epilepsy Clinic tested construct validity.

3. Results

3.1. Frequencies of the items

The questionnaire as developed contained the following variables with range 1 through 4 (Table 1).

Skewness and kurtosis analysis showed normality of the variables of the questionnaire except one item ('I think that my PCP considers me hard to deal with') that showed positive skewness (2.44; standard error of skewness .19) as well as positive kurtosis (5.88; standard error of kurtosis .38). All patients disagreed with this item. Therefore, this item was removed from the questionnaire and further analysis [25].

3.2. Factor analysis

Factor analysis (principal component with varimax rotation) showed two factors (Table 2).

The Eigen value of the first factor is 7.509. This accounts for 58% of the total variance explained. The Eigen value of the second factor is 1.198. This accounts for 9% of the total variance explained.

The item 'I benefit from the treatment of my PCP', loads too high on both factors to be useful into a factor model. In the two factorial model, a variable should preferably load $>.40$ in one factor and $<.30$ in the other factor, according to Nunnally [26]. This item was out of that range and therefore eliminated. These factors have face validity and suggest different aspects of the patient-doctor relationship.

Table 2
Factor analysis. Rotated component matrix.

Item PDRQ	Factor 1	Factor 2
My PCP understands me	.884	
I trust my PCP	.880	
My PCP is dedicated to help me	.849	
I can talk to my PCP	.847	
I feel content with my PCP's treatment	.835	.331
My PCP helps me	.778	
My PCP has enough time for me	.686	.312
I benefit from the treatment of my PCP	.638	.523
My PCP and I agree on the nature of my medical symptoms	.628	.391
I find my PCP easily accessible	.489	
Thanks to my PCP I feel better		.809
Thanks to my PCP I gained new insight		.770
I can handle my medical symptoms now (Even if my PCP and I have no further meetings)		.654
My medical symptoms will probably disappear		.435

The first factor focuses on the empathic style and availability of the doctor. The second factor focuses on the medical symptoms of the patients.

3.3. Reliability

The first factor of the questionnaire, consisting of nine variables, had a Cronbach's alpha of .94, which delineates good internal consistency of this subscale. If more items of the first factor are eliminated, Cronbach's alpha drops under .90. The second factor of the questionnaire, consisting of four variables, had a Cronbach's alpha of .68, for 4 items still acceptable.

3.4. Test-retest reliability

Of the 110 primary care setting patients, 35 patients filled out the questionnaire for a second time. Mean follow-up time was 61 days. Sex was 64% female in the test-retest group as well in the nonparticipants. Mean age was 46 years in the test-retest group and 44 years in the nonparticipants group. The mean PDRQ score in the nonparticipants was

3.66 (SD 1.07). Paired *t* tests were performed. The mean PDRQ score was 3.84 (SD .68); the mean PDRQ score after 2 months was 3.63 (SD .80), thus remaining approximately the same. Pearson *r* is .61, and significant. This is a reasonable test-retest reliability given the long follow-up period of 2 months.

3.5. Construct validity

To test the ability of the PDRQ to discern difference between subsamples, the scores of the primary care setting patients were compared with those from the patients from the Epilepsy Clinic by *t* test. One-tail probabilities are reported in Table 3. All comparisons have shown significant differences in the PDRQ scores in the 2 different groups.

3.6. Elimination of items

As noted before, the choice for items related to medical symptoms was inspired by the HAQ, in which problem-related items exist. In the HAQ of Luborsky, opinions of the patients about their psychological problems could be seen as related with the psychotherapeutic relationship. As indicated by the results, in the primary care setting, opinions of the patients about their medical symptoms cannot be extrapolated to the doctor-patient relationship in the same way. Patients in a primary care setting often have medical symptoms because of the discomfort of physical illness, such as diabetes, hypertension, cardiac problems and so on. On the one hand, they may suffer from the medical symptoms, and they may (realistically) think that the symptoms will not disappear. On the other hand, they may still be very satisfied about the doctor-patient relationship with their PCP. The items in the second factor have in common that they assume a change in symptoms because of the effort of the PCP, an assumption that turns out to be incorrect in this primary care population. Considering the 2-factorial structure of the PDRQ, we want to focus on the nine items clearly assessing the patient-doctor relationship with a focus on empathic style and availability of the doctor. Also, the internal consistency of the first factor is high, and of the second moderate. Therefore, we eliminate the second factor. This leaves us with a 9-item questionnaire.

Table 3
T-test comparing primary care setting with Epilepsy Clinic

	Setting	N	Mean	SD	T	Df	Significance (one-tailed)	Mean difference
PDRQ factor 1	PC	105	4.06	.77	3.12	158	.001	.46
	Epilepsy Clinic	55	3.60	1.09				
PDRQ factor 2	PC	103	3.17	.83	5.02	155	.000	.73
	Epilepsy Clinic	54	2.44	.91				
PDRQ total	PC	101	3.78	.71	3.94	153	.000	.53
	Epilepsy Clinic	54	3.25	.94				

4. Discussion

We intended to develop a short questionnaire, fit for quantification of the patient-doctor relationship from the perspective of the patient, with the focus on the helping attitude of the doctor. The PDRQ was developed with the Dutch validated version of the HAQ as starting point. Our initial 15-item questionnaire was reduced to 9 items as a result of further analysis.

The total response rate (65%) as well as the response rate in the primary care setting (55%) compare favorably with the response rate of Poulton [27] in a similar population (46%), but are still moderate. One reason for this could be the fact that the patients had to send their questionnaire to the researcher by mail. There were no indications for a ceiling effect. The questionnaire turned out to be highly reliable, as far as internal consistency is concerned.

It is unfortunate that the test-retest response rate was not higher than 33%. Nevertheless, we were able to make an estimation of the test-retest reliability.

There were no significant differences in terms of demographic variables and initial PDRQ-9 scores between the 35 participants in the test-retest study and the 130 nonparticipants. Even lower response rates can be acceptable. For example, Goodger et al. [28] found a test-retest response rate of 20% in the assessment of a short social support scale in a similar population.

To investigate the ability of the instrument to discriminate between sub-samples with different patient-doctor relationships, we compared scores of this questionnaire in 2 different settings, namely the primary care setting and an Epilepsy Clinic. The assumption was that patients, who were assessed in a supraregional specialist clinic for difficult to diagnose or treatment refractive epilepsy, or for medically unexplained symptoms mimicking epilepsy, might have a more negative view on their PCP. It could be possible that the patients would give a more honest opinion of their PCPs, also if this would mean a negative evaluation, as the data were collected away from the primary care practice, or that they contrasted the former consultations with their PCP with those they experienced in the Epilepsy Clinic. Though these are speculative reasons, we wanted to explore if this would show up in the questionnaire ratings. Because lower ratings were detectable in the epilepsy group, support for discriminatory power was found. Though no other measures were used that assess the doctor-patient relationship, these findings suggest that the PDRQ-9 can discriminate between respondents from a specialist clinic and normal controls.

Of course, when evaluating a questionnaire that focuses on the patient's perception of the helping attitude of his PCP, one should be aware of the possibility that patients will give a socially acceptable answer. We tried to eliminate this problem by assuring patients anonymity, and by collecting the questionnaires by direct mail to the psychiatry department. Still, it remains possible that patients were

biased towards a positive judgement in the assessment of their PCP. However, the mean scores do not suggest that, because they do not linger in the upper region but rather in the middle. Also, patients suspected of a less positive doctor-patient relationship did indeed give a statistically significant less positive opinion. This suggests that the PDRQ-9 might be able to discriminate between good and moderate doctor-patient relationships.

A difference between the PDRQ-9 and a patient satisfaction questionnaire, is the choice not to focus on quality of care, technical items, or communication patterns. Rather, it focuses on the helping attitude of the PCP as experienced by the patient, as it was our intention to make a questionnaire derived from the concept of the helping alliance, a psychotherapeutic concept, adapted for the use in general practice. However, it could be argued that there is overlap. For example, 4 of the MOS-9 items ask patients their satisfaction with: (1) 'The thoroughness, carefulness, and competence of the doctor you saw'; (2) 'Time spent with the doctor you saw'; (3) 'Explanation of what was done for you'; (4) 'The personal manner (courtesy, respect, friendliness, sensitivity) of the doctor you saw'. Therefore, future research with the PDRQ-9 should also assess how it is similar and different from standard patient satisfaction questionnaires.

The evidence for reliability and validity of the PDRQ-9 is promising but further validation studies will be needed to confirm this finding. There are measures to assess the patient-doctor relationship from the doctor's standpoint. One of the best validated is the 10-item Difficult Doctor Patient Relationship Questionnaire [3]. Subsequent studies might study correlations between patients' assessment of the relationship and doctors' assessments.

The primary care sample only included 5 PCPs, and subsequent studies should sample a larger number of PCPs from different types of primary care settings, such as large practices, small practices, managed care, as well as disciplines, such as family practice, PCPs, and general internal medicine.

Not only should this questionnaire test performance be validated in other settings and populations, but also should be explored whether it correlates with clinically important outcomes. For example, do doctors with higher scores on this questionnaire have higher follow up rates, are the patients more likely to comply, do they have higher rates of diagnosis of mental disorders? Are high scores on this questionnaire correlated with better outcomes in terms of morbidity of diseases? Does the instrument have discriminatory power in terms of disease burden, functional status or mental disorder in patients?

The results encourage its further validation and use in primary care and public health research as well as in health promotion strategies. The PDRQ provides researchers with the opportunity to use a brief measure of the doctor-patient relationship from the patient's perspective, which has been assessed specifically in primary care. The patient-doctor

relationship as a derivative of the therapeutic alliance may be monitored with the new questionnaire, thus representing a valuable tool for scientific and practical purposes.

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APPENDIX PDRQ-9

Instruction:

You will read nine statements that a person can make about his/her PCP. Please choose the appropriateness of each statement for your PCP by marking one number per statement. The meaning of the numbers is as follows:

- 1 = not at all appropriate
- 2 = somewhat appropriate
- 3 = appropriate
- 4 = mostly appropriate
- 5 = totally appropriate

1	My PCP helps me	1	2	3	4	5
2	My PCP has enough time for me	1	2	3	4	5
3	I trust my PCP	1	2	3	4	5
4	My PCP understands me	1	2	3	4	5
5	My PCP is dedicated to help me	1	2	3	4	5
6	My PCP and I agree on the nature of my medical symptoms	1	2	3	4	5
7	I can talk to my PCP	1	2	3	4	5
8	I feel content with my PCP's treatment	1	2	3	4	5
9	I find my PCP easily accessible	1	2	3	4	5

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